$f(x) = \frac{x^2+5x+6}{x+2}$ ← $x+2=0$, $x=-2$

$f(-2) = \frac{(-2)^2+5(-2)+6}{-2+2} = \frac{0}{0}$

Discont at $x=-2$, b/c $f(-2)$ DNE.

9.2.039. EIT

$$y = \frac{33}{(p+7)^{2.5}}$$

1) $p+7=0$, $p=-7$ $\quad y(-7) = \frac{33}{(-7+7)^{2.5}} = \frac{33}{0}$ not cont.

2) Cont at $p=24$?

$\lim_{p\to 24} y$

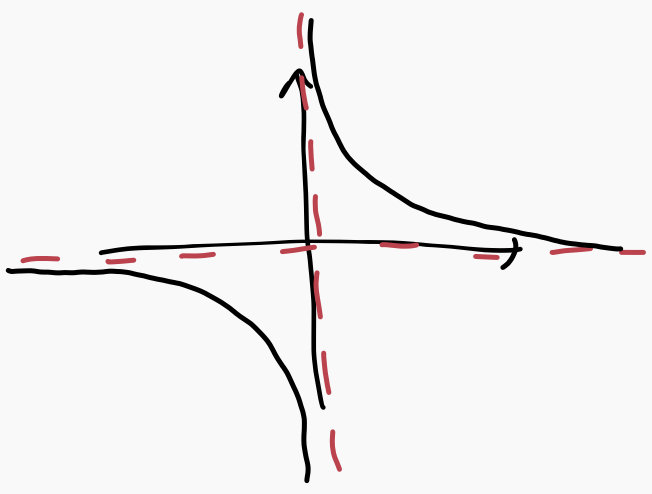
| p | y |
|---|---|
| 23.9 | |
| 23.99 | |
| 23.999 | |
| 23.9999 | |

| p | |
|---|---|
| 24.1 | |
| 24.01 | |
| 24.001 | |
| 24.0001 | |

Ex $y = \frac{1}{x}$

| x | $\frac{1}{x}$ |
|---|---|
| -10 | -.1 |
| -100 | -.01 |
| -1000 | -.001 |
| -10000 | -.0001 |

| x | $\frac{1}{x}$ |
|---|---|
| 10 | .1 |
| 100 | .01 |
| 1000 | .001 |
| 10000 | .0001 |



$\lim_{x\to-\infty} \frac{1}{x} = 0$ $\quad \lim_{x\to\infty} \frac{1}{x} = 0$

Ex $y = \frac{1}{x^2}$

$\lim_{x\to-\infty} \frac{1}{x^2}$

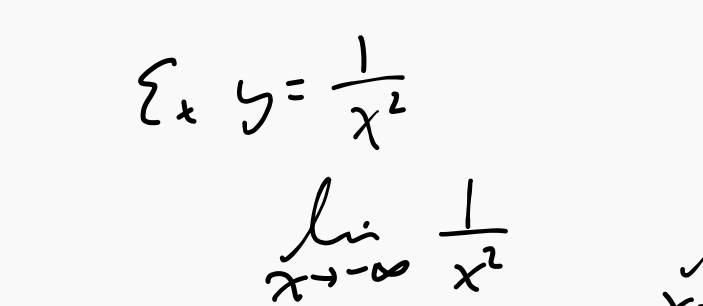
| x | $\frac{1}{x^2}$ |
|---|---|
| -10 | |
| -100 | |
| -1000 | |
| -10000 | |

$\lim_{x\to\infty} \frac{1}{x^2}$

| x | $\frac{1}{x^2}$ |
|---|---|
| 10 | |
| 100 | |
| 1000 | |
| 10000 | |

B.1 curve

a) $f(x)=3$ over $x=0 \to x=2$



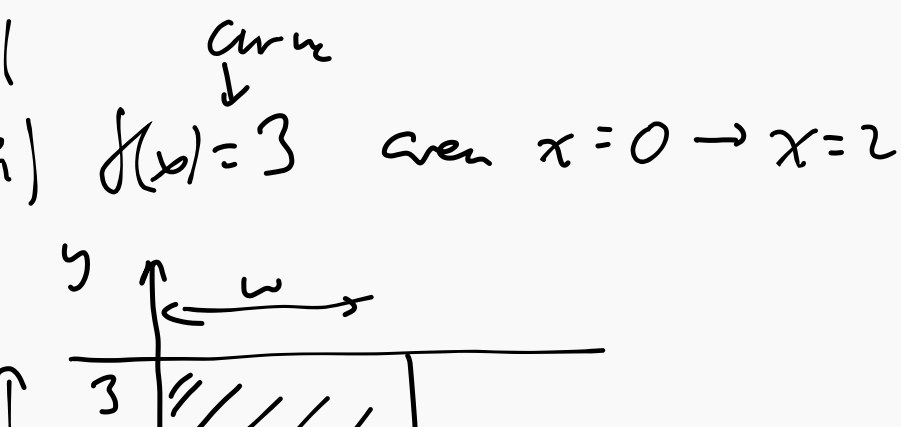

$A = h \cdot w$
$= 3(2) = 6$

b) $f(x)=x$

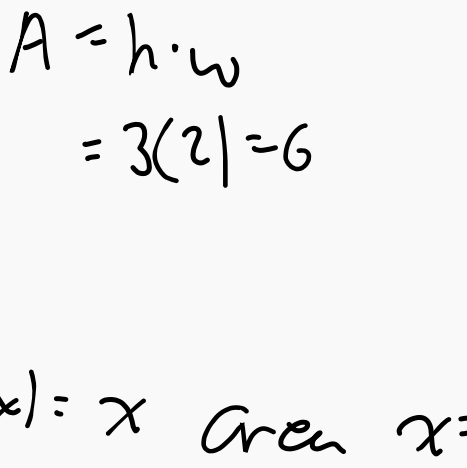



$A = \frac{1}{2}bh$
$= \frac{1}{2}(2)(2)$
$= 2$

d) $f(x)=x$ over $x=1 \to x=3$




c) $f(x) = x^2$, $x:0 \to x:2$




$$\int_0^2 x^2\,dx = \frac{2^{2+1}}{2+1} - \frac{0^{2+1}}{2+1}$$
$$= \frac{2^3}{3} - 0 = \frac{8}{3}$$